

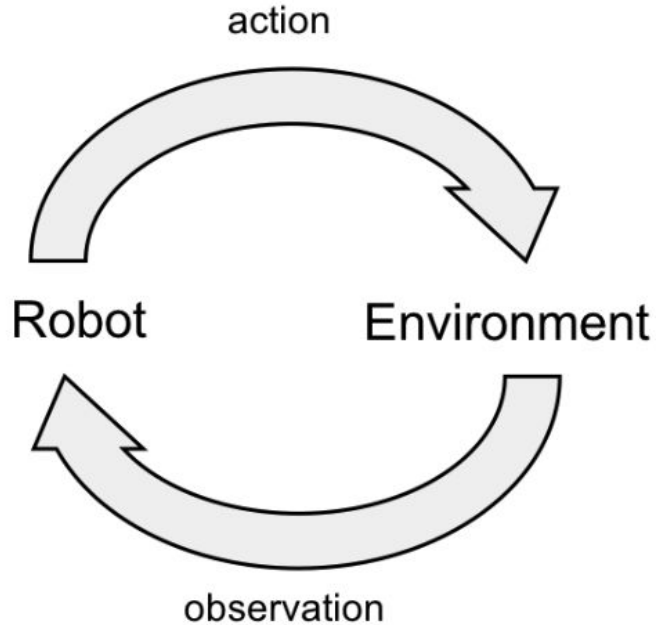
Autonomous Systems

Week 04: Embodiment in Autonomous Systems

Murat Kirtay

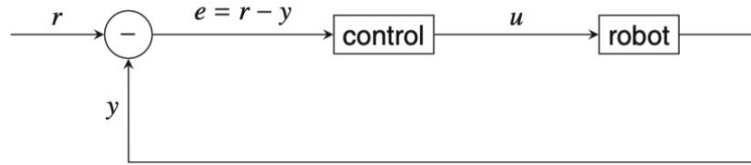
<http://www.crossvalidate.me/>

Previously in Autonomous Systems



Questions related to previous sessions

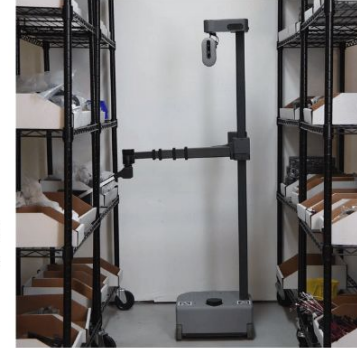
Closed loop control system (feedback control)



$r \rightarrow$ **reference value**: the specification of robot's task, e.g., the position of the robot relative to a stack of shelves, and the distance of gripper arm from object to be picked up.

$y \rightarrow$ **output**: the actual state of the robot, e.g., the distance to an object.

$u \rightarrow$ **control value**: robots can not use r value directly, if the r is the position of the robot relative to a shelf, the u will be power settings of the motors on the robot.

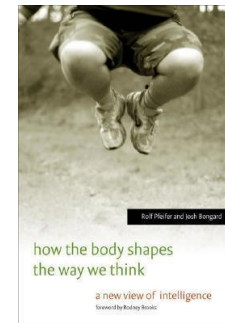
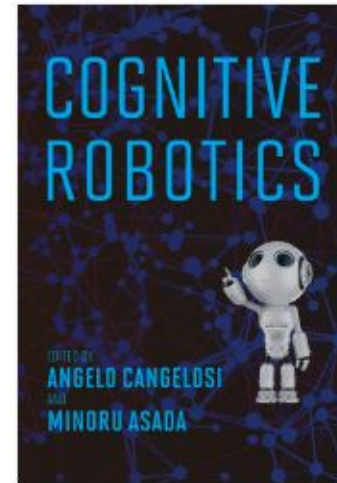
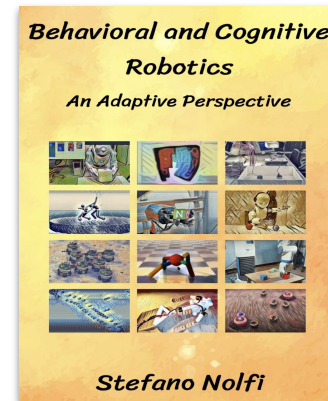


<https://hello-robot.com/>

ROBOT PRIMITIVES	INPUT	OUTPUT
SENSE	Sensor data	Sensed information
PLAN	Information (sensed and/or cognitive)	Directives
ACT	Sensed information or directives	Actuator commands

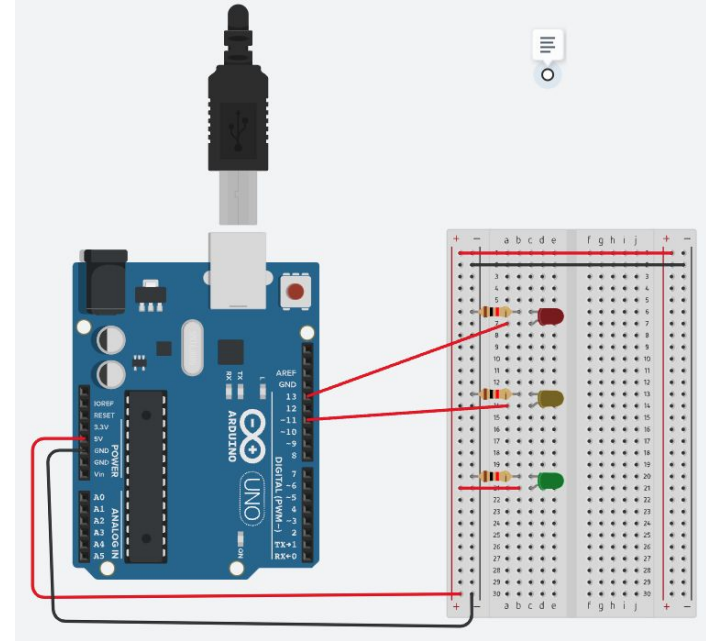
Weeks	Lecture topics
01	Intro to Autonomous Systems & Course Logistics
02	Robotics: Sensors and Actuators. & Braitenberg Vehicles
03	Controllers (Brains) in Autonomous Systems
04	Embodiment in Autonomous systems
05	Reinforcement Learning in Autonomous Systems
06	Unsupervised Learning in Autonomous Systems
07	Multimodal Learning, Interaction, and Communication
08	Affective and social Human-Robot and Robot-Robot Interactions
09	Cognitive Robotics
10	Simulation environments for autonomous systems
11	Selected topics in autonomous systems
12	Invited Speaker I
13	Invited Speaker II

Today's Agenda



Practice session

- Bring your Arduino Kits
- We ordered more Kits, you can take a kit if you do not have one.
- We will do
 - LED with Arduino in TinkerCad
 - LED with actual Arduino
- Please install Arduino IDE



We will also have open buffet of electronics that you can use various circuit components and tools.

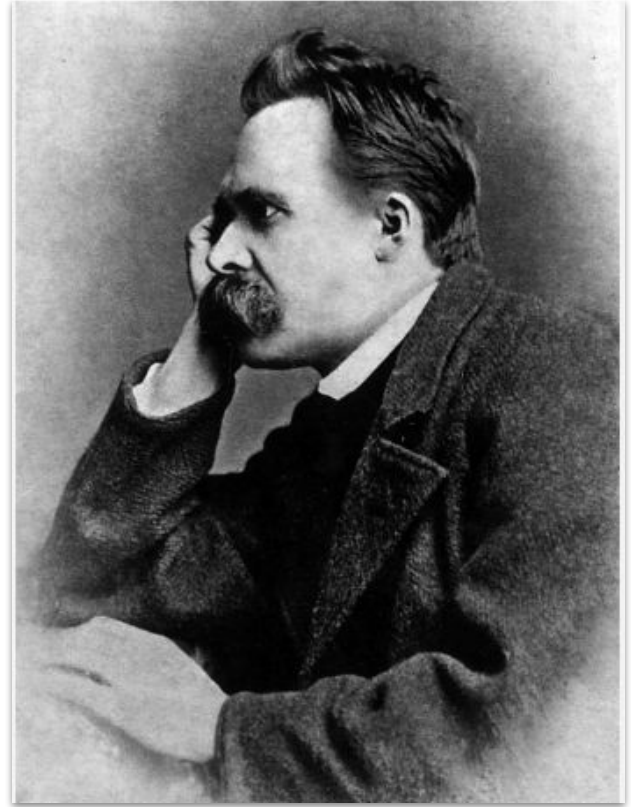
Mandatory quote for our lecture

“I think, therefore
I am”



Mandatory quote for our lecture

"There is more intelligence in your body than in your best wisdom."



Today's questions

Does intelligence require a body?

The growing discipline of embodied cognition suggests that to understand the world, we must experience the world

Katrin Weigmann

- 1- Do we need a body to be considered as an intelligent agent?
- 2- What is the role of having a body in cognition/emotion?
- 3- What is it like to be a robot?
- 4- Where body ends? Where external world starts?
- 5- Can you have a sense of self without having a body?

What is embodiment?

- The term **embodiment** is used to indicate agents that possess a physical body.
- The body is composed of parts made of different materials and arranged in a given **morphology**
- Morphology constraints the behavior of the robot.

Human morphology



Animal morphology



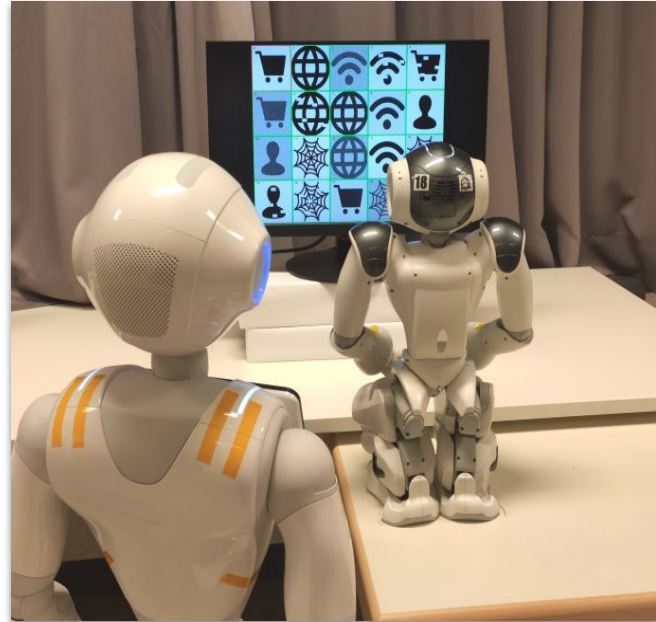
A service robot



Exam Questions

Do you think ChatGPT is an embodied agent?, Explain why?

Do you think Pepper and Nao robots are embodied agent?, Explain Why?

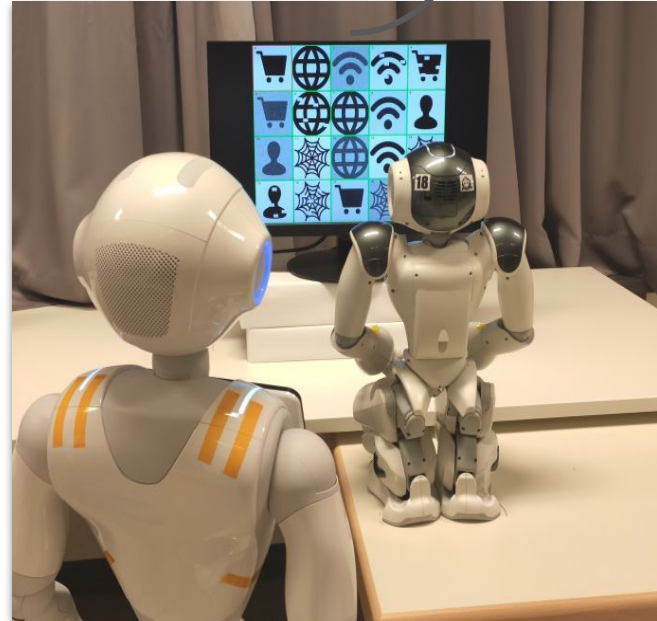
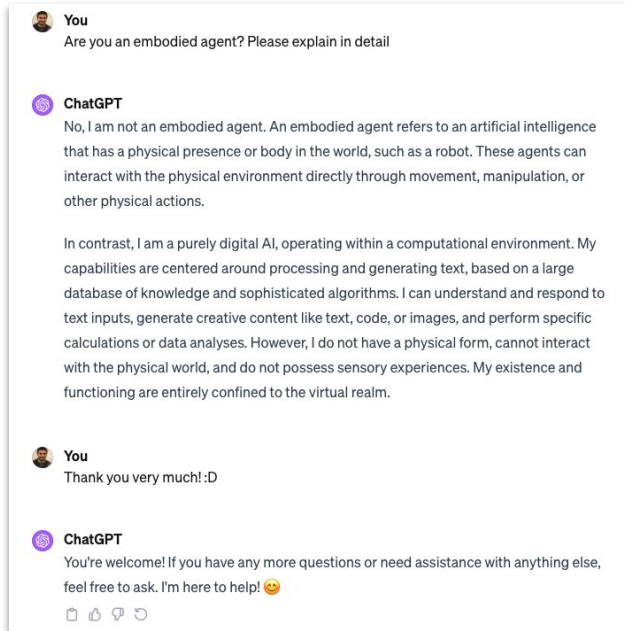


Questions

Do you think ChatGPT is an embodied agent?

Do you think Pepper and Nao robots are embodied agent?

YES.



Recall that dead fish can swim and brainless robot can walk



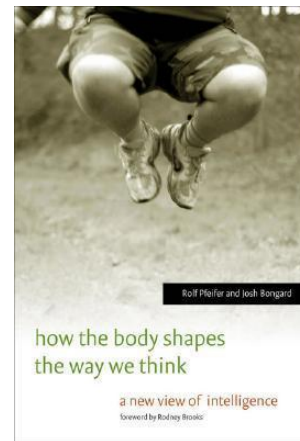
But how?

Embodied Intelligence/Cognition

(a)

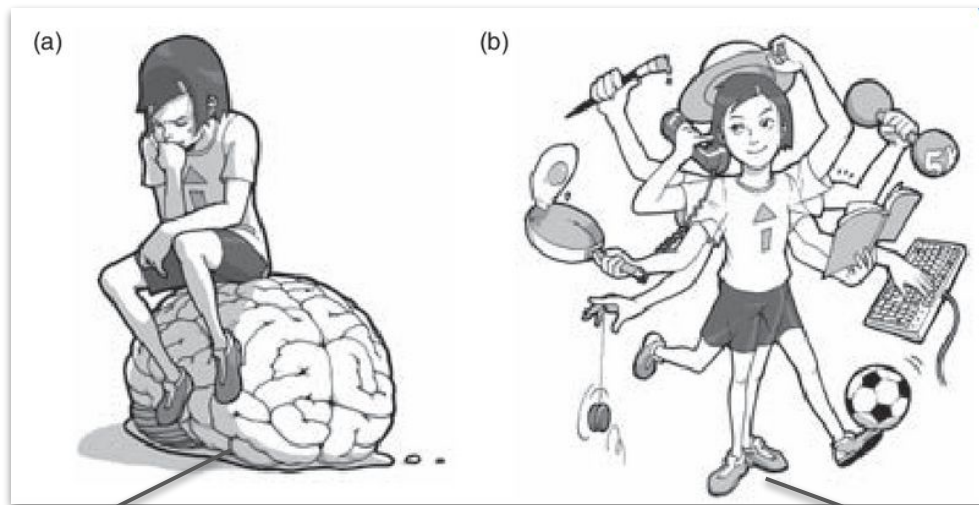


(b)



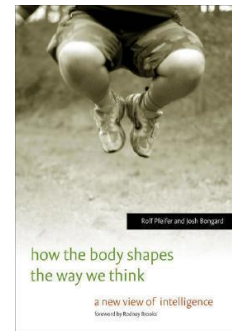
Two ways of approaching intelligence. **(a) The classical approach.** The focus is on the brain and central processing. **(b) The modern approach.** The focus is on the interaction with the environment. Cognition is emergent from the system-environment interaction,...

Classical vs. Modern approach to intelligence

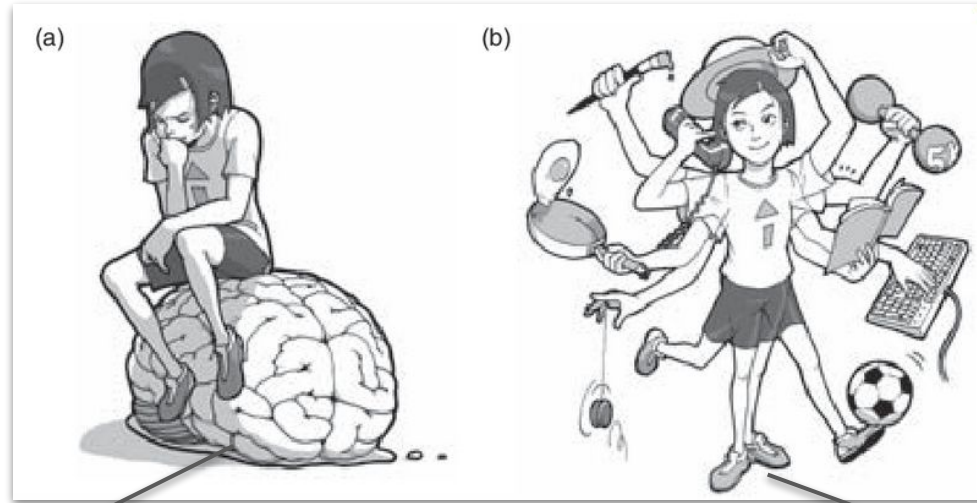


The classical approach. The focus is on the brain and central processing.

The modern approach. The focus is on the interaction with the environment.



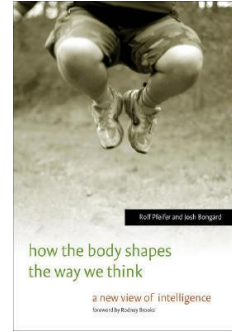
Classical vs. Modern approach to intelligence



The classical approach. The focus is on the brain and central processing.

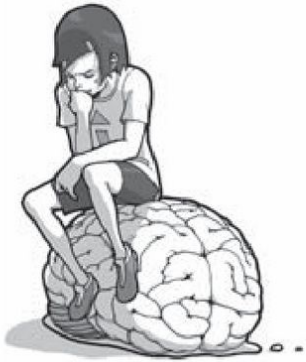
The modern approach. The focus is on the interaction with the environment.

Three components of embodied intelligence: BRAIN, BODY, and ENVIRONMENT.



Embodied Intelligence/Robotics

(a)

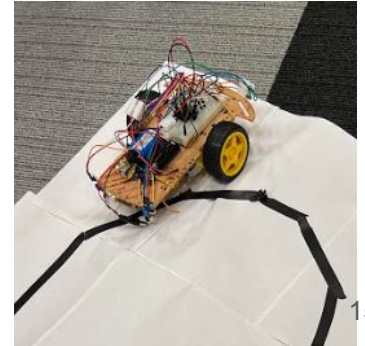
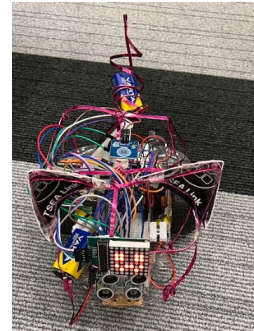


(b)



“Intelligence cannot merely exist in the form of an abstract algorithm but requires a physical instantiation, a body.”
Pfeifer and Scheier.

That is why you will build an embodied autonomous agent for your project.

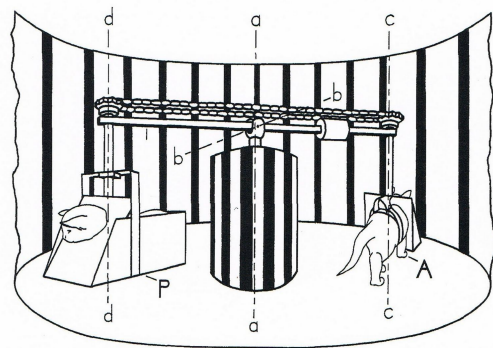


A thought experiment



What will happen if you embed the same control algorithm on the robots with different bodies?

The role of body (and movement) in cognition

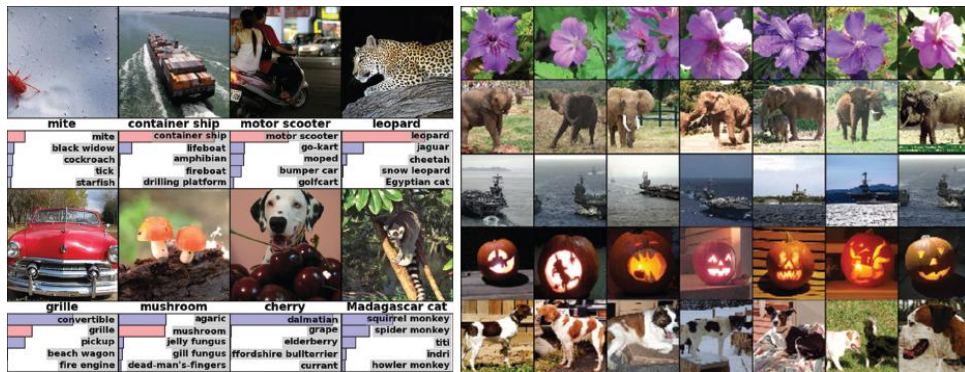


The **passive kitten** received visual stimulation identical to that of **the active** kitten but did not have the opportunity to move by itself; instead, its movement was dependent on the active kitten's actions.

- Active kittens developed normal visual perception skills.
- Passive kittens showed significant deficits in visual perception.

“Intelligence cannot merely exist in the form of an abstract algorithm but requires a physical instantiation, a body.”
Pfeifer and Scheier.

ImageNet vs. Infants | Dis-embodied vs. embodied



- More than 15 million images high resolution images
- **Targets:** 2200 categories,
Features: mean centered RGB matrices with a size of 256x256
→ learned filters by CNNs



“... after the **second birthday**. At that point children **can infer the extension of a whole category from one example**. Given just one instance of a novel category, and its name, 2-year-old children immediately generalize that name in an adult-like manner.”

Why robots should be embodied?

“Embodied (robotic) models that share some bodily and sensorimotor features with the organism they are supposed to model might be preferable to **purely computational** or **mathematical model**.“



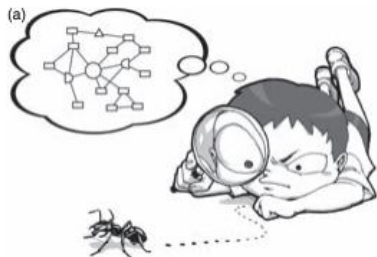
- To understand/model how human affordances grounded with interaction among object, tool, and action.
- Use a humanoid robot with hands to model the sensorimotor capability of humans.

How to build an embodied agent?

Synthetic methodology (understanding by building): if you want to understand human walking, we must build an actual walking robot.



How to build an embodied agent?

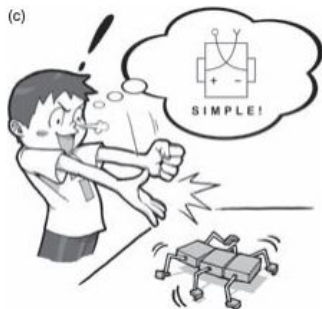


The synthetic methodology: scientist trying to understand ant behavior.



(a) Directly mapping observed behavior onto an internal representation.

(b) Using this representation to control walking in a robot fails.



(c) Applying a much simpler model—resembling a Braitenberg architecture—to control walking in a robot.

This latter, simpler model seems more plausible as an explanation of the originally observed behavior.

Embodied, simulated, and embodied non-collocated.

Embodiment and Human-Robot Interaction: A Task-Based Perspective

- (a) The co-located physical robot condition, as a typical model of human-robot interaction. The robot is placed on the table in front of the participant.
- (b) The remote presence robot case, in which the audio-video tele-conferencing system provides real-time playback of the robot, despite being situated in a different room.
- (c) The simulated robot case. The same screen and audio setup as (b) is used, but without a physical robot; instead a simulated robot world is shown.



Fig. 1 **Left**, the simulated robot used for the embodiment study; **center**, the embodied non-co-located robot; **right**, the embodied co-located robot. In this study, a robot was interacting with a user autonomously during a game-playing task. The users were asked to rate the robots for intelligence, watchfulness, and likability [60]. The users rated the embodied robot highest on all these ratings, followed by the embodied non-co-located robot, followed by the virtual agent. These results support the use of an embodied agent over a virtual agent on a screen

Left and right description are mixed in the figure caption.

How many bodies we (or robots) have?

- **the social body,**
as it appears to others,
- **the sensorimotor body,**
which interacts with the environment,
- **the living body,**
which has to self-regulate and self-maintain,
- **the lived body,**
as it is experienced by an agent itself.



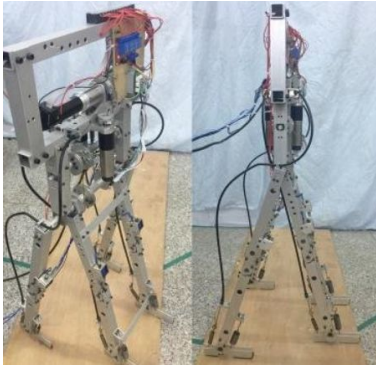
In class philosophical discussion

“a robot with bodily structures, interaction patterns and development similar to those of human beings ... could possibly recapitulate [human] ontogenesis, leading to emergence of intentionality, consciousness and meaning.”

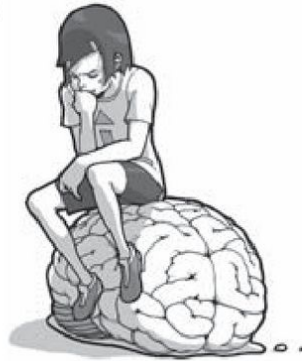
Jordan Zlatev.

Morphological computation

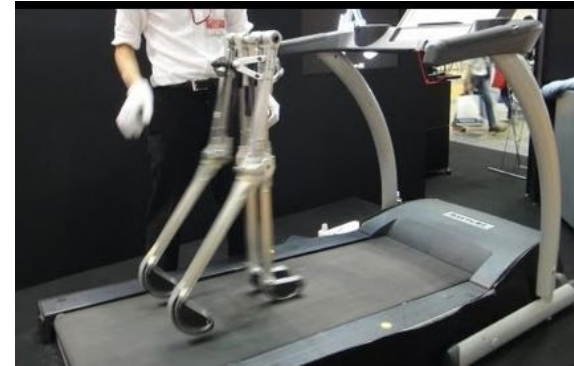
Morphological computation: the ability of body to perform the same function of the brain.
“**offloading computation from the brain to the body,**”



(a)



(b)

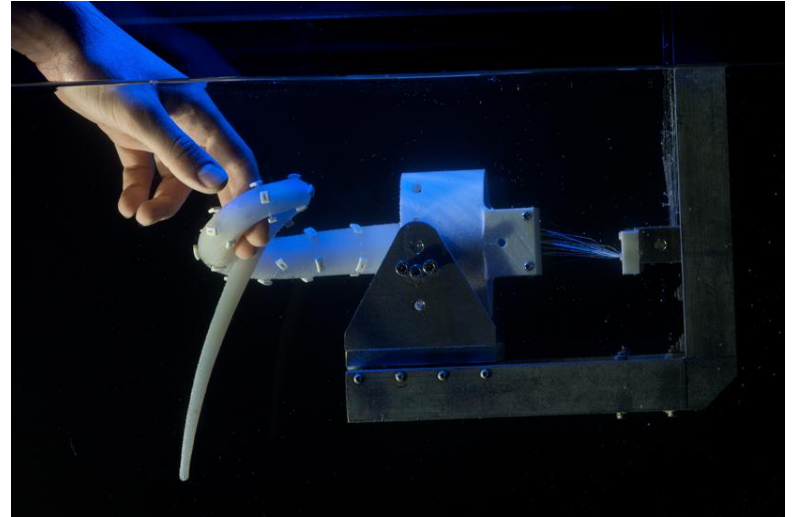
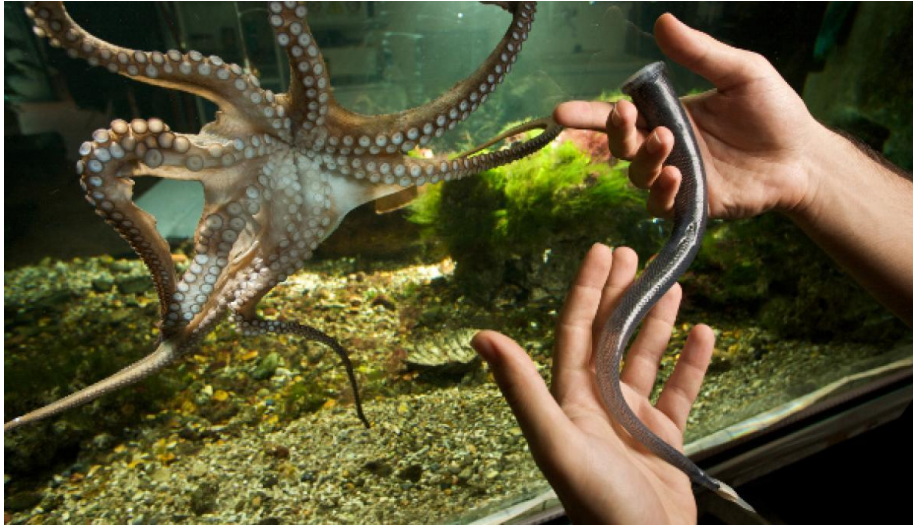


- Brain co-determines the actions performed by the robot
[quasi passive walkers walking on flat terrains]
- Body co-determines the actions performed by the robot
[passive walkers]

Morphological computation in animals and robots



“the use of soft materials and continuum deformations and stiffening make control more difficult in computational terms using traditional model-based robot control approaches. Therefore, **the robots of tomorrow will benefit from the interconnection and interaction between morphological computation and soft robotics.**”



Passive walkers are brainless, so are they robots?

- body
- sensors
- actuators
- **brain**
[controller]
- environment
- function



Passive walkers are brainless, so are they robots?

- **body**
- sensors
- actuators
- **brain**
[controller]
- environment
- function



Yes, they are robot because the body performs the same function of the brain.

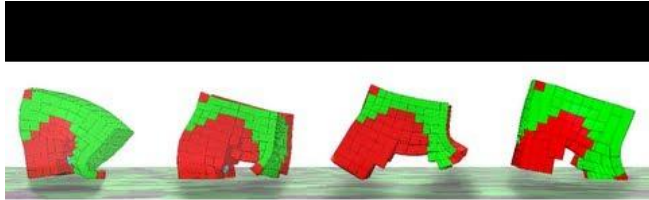
Advantages of morphological computation

- the robot can use morphological computation to perform regulations that would be difficult or impossible to perform through the brain
- robots can produce “complex” behavior with “simple” brains, or without having a brain.

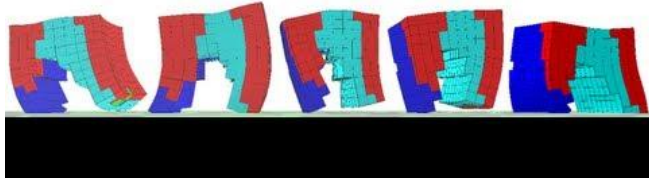
Implications

- that the functions played by the body and by the brain are not separated
- the role played by the brain cannot be explained without considering the role of the body and vice versa.

Body adaptation- evolution.



Evolution in Action!



Results: The evolutionary process produces robots with more and more adapted bodies capable to walk better and better

Goal: cubic cells made of different materials to move as fast as possible over a planar surface.

Body: cells are distributed within a grid of 10x10x10 voxels, i.e., 3D pixels.

a bone cell made of rigid material,

a tissue cell made of soft material,

a muscle cell that periodically contracts and expands

- **Sensorless** (except for local expansion/contraction)
- **Brainless:** ability to walk comes exclusively from their morphological characteristics.

Xenobots

A scalable pipeline for designing reconfigurable organisms



passive and contractile cells **evolved in simulation** and **then manufactured by using living [brainless] cells.**

- evolving in simulation creatures formed by rigid and contractile cells for the ability to swim in a liquid environment
- creating living agents formed by stem cells assembled to create morphologies similar to those of the evolved creatures

Results:

- a) these creatures display behaviors similar to that displayed by the corresponding evolved creatures in simulation
- b) they are capable to explore their aqueous environment for days **without human intervention.**

We saw the evolution of **brainless** biological
and artificial creatures.

What happens if you allow brain and body
co-evolve together?

Co-evolving the body and the brain

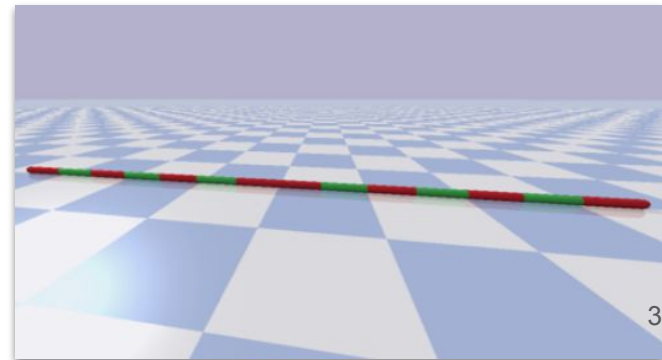
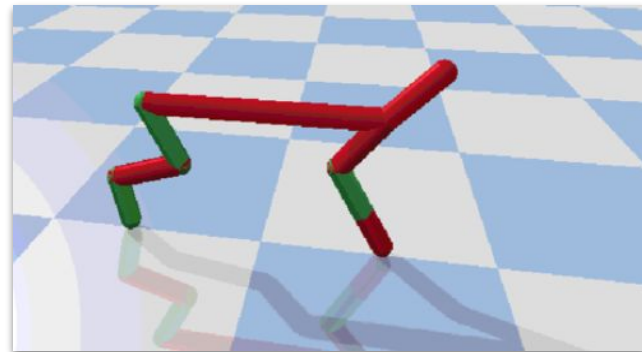
Hand designed morphology vs. brain/body co-evolving morphology.

Body

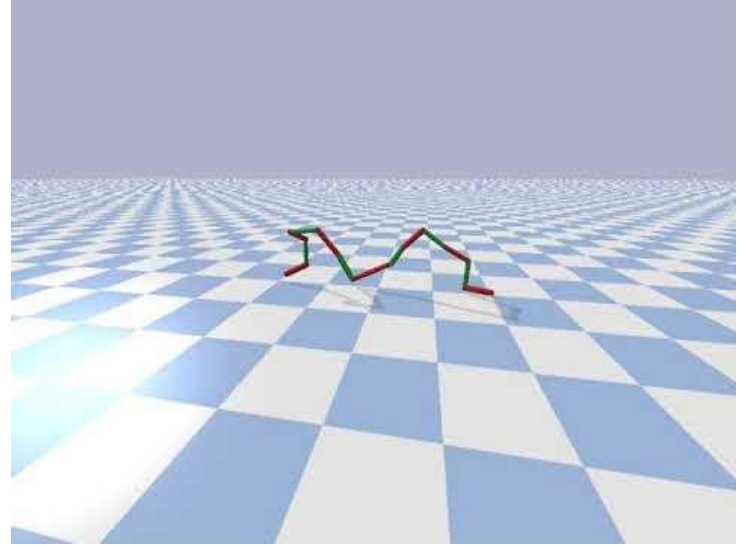
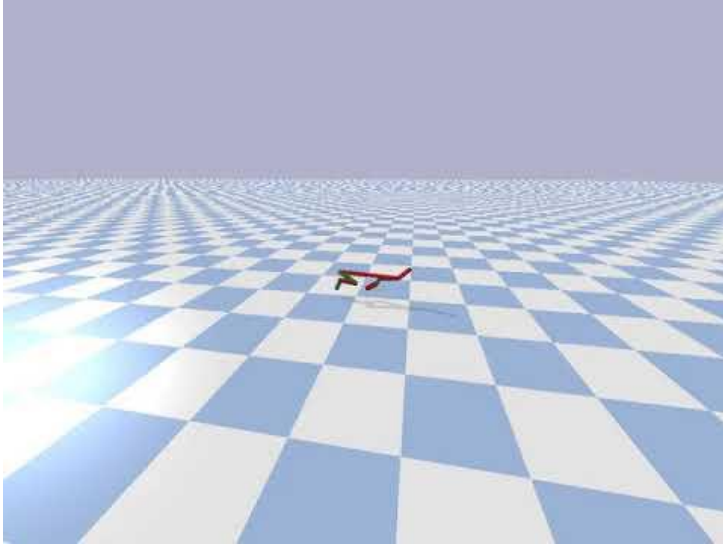
- Half-Cheetah robot's of: cylindrical elements connected with hinge joints.
- The general bauplan: # 11 cylindrical elements

Brain

- Feed-forward neural network
- Sensory neurons
 - Contact sensors
 - Velocity and orientation of segment
 - Angular position of joints
- Motor neurons
 - The torque applied to the actuated joints



Results

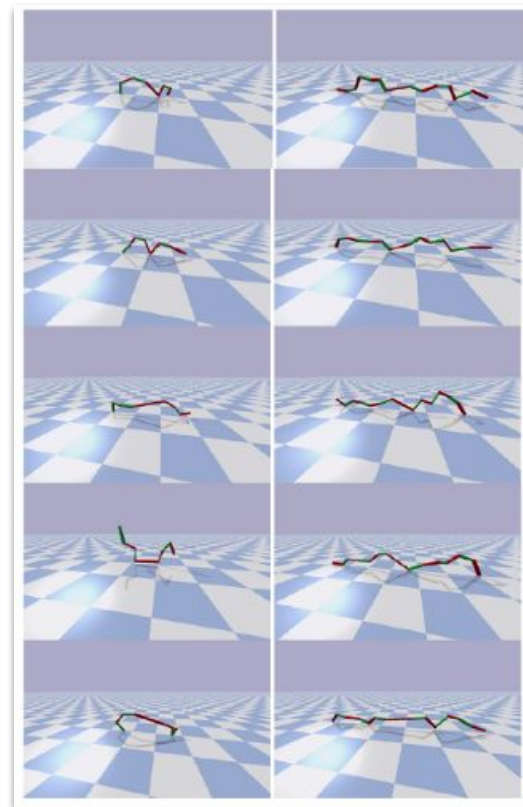


- Evolving brain-body model achieves better results
- Evolution and discover suitable morphology to achieve locomotion

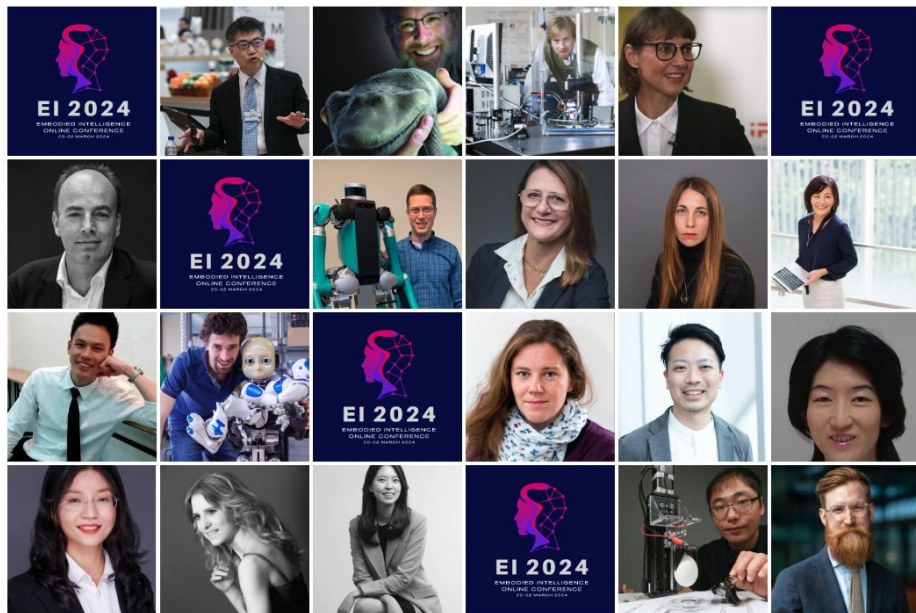
Co-evolving the body and the brain

We assume that the ideal approach to body and brain co-evolution should be

- **Simple**
[set of elementary elements, analogous to amino acids]
- **Should have high expressive power**
[generation of wide spectrum phenotypes, speciation by natural selection]
- **Should enable the generation of evolvable solutions**
[generation of agent with genetic variations]



Embodied Intelligence [online conference]



- What is (embodied) intelligence?
- How does computation come about from embodiment?
- How does language emerge?
- How do we understand intelligence and what theories do we have to represent it?
- How can we measure complexity?
- What can not be done by computational intelligence?
- What ethical questions arise from EI?

An online conference on Embodied Intelligence.

- Please consider to register; it is free.
- You will experience cutting-edge research and interact with researchers.

Introduction to Embodied AI by J. Bongard in 5 minutes

